# Endpoint Hardy Membership and a Logarithmic Mismatch Barrier

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**Abstract**

At the deterministic coefficient radius $\rho_* = 1.4267874838\ldots$, the actual complement-to-anchor mismatch has a sharp qualitative split: it belongs to $H^2(\rho_*)$ for every fixed noise, but it never belongs to $H^\infty(\rho_*)$. Endpoint $H^2$ convergence would still close the weighted prefix at $R = 1.4$, with conversion constant $4.992111\ldots$. However the exact odd anchors and the complement mass cap force an endpoint norm lower bound of order $1/\sqrt{\log M_\sigma}$, hence at least order $1/\sqrt{\log(1/\sigma)}$. This barrier tends to zero, so it does not prove endpoint convergence or nonconvergence.

## 1 Endpoint functions

Let

$$\rho_* = q_*^{-1} = 0.85\lambda, \qquad q = \frac{1}{2}, \qquad g_\sigma(z) = \sum_{n\geq 2} \frac{\tau_{\sigma,n} - a_n}{n} z^n.$$

For the modulus complement, $|\tau_{\sigma,n}| \leq M_\sigma q^{n-2}$ with finite $M_\sigma \leq \sigma^{-1}$. The deterministic anchors satisfy $|a_n| < 48 q_*^n$ and, for odd $n$,

$$a_n = \frac{q_*^n}{1 + \lambda^{-n}}.$$

## 2 Membership theorem

**Theorem 2.1.** *For every fixed $\sigma > 0$,*

$$g_\sigma \in H^2(\rho_*), \qquad g_\sigma \notin H^\infty(\rho_*).$$

*Proof.* The noisy logarithm is bounded at $\rho_*$ because $q\rho_* = q/q_* < 1$ and its coefficients have the geometric mass envelope. The target logarithm belongs to $H^2(\rho_*)$ because

$$\sum_{n\geq 2} \frac{|a_n|^2 \rho_*^{2n}}{n^2} \leq 48^2 \sum_{n\geq 2} \frac{1}{n^2} < \infty.$$

Thus their difference belongs to $H^2$.

The odd part of the target logarithm has positive radial values

$$\sum_{\substack{n\geq 3 \\ n \text{ odd}}} \frac{r^n}{n(1+\lambda^{-n})}, \qquad 0 < r < 1,$$

which diverge as $r \uparrow 1$. Hence the target is not bounded on the endpoint disk. Subtracting the bounded noisy logarithm cannot make it bounded, proving the $H^\infty$ exclusion. $\square$

## 3 Endpoint conversion and lower barrier

Put $R = 1.4$ and $x = R/\rho_*$. Every endpoint $H^2$ mismatch satisfies

$$\sum_{n\geq 2} \frac{|\tau_{\sigma,n} - a_n| R^n}{n} \leq \|g_\sigma\|_{H^2(\rho_*)} \frac{x^2}{\sqrt{1-x^2}},$$

where

$$\frac{x^2}{\sqrt{1-x^2}} = 4.992111068649647\ldots.$$

**Theorem 3.1.** *Let $M > 0$ and let $N(M)$ be the least odd $N \geq 3$ satisfying*

$$Mq^{N-2} \leq \frac{1}{4} q_*^N.$$

*Then every modulus-capped complement of squared mass $M$ obeys*

$$\boxed{\|g\|_{H^2(\rho_*)} \geq \frac{1}{\sqrt{32N(M)}}}.$$

*Consequently, for constants $C, M_0 > 0$,*

$$\|g\|_{H^2(\rho_*)} \geq \frac{C}{\sqrt{\log(e+M)}} \qquad (M \geq M_0).$$

*For the actual complement mass $M_\sigma \leq \sigma^{-1}$,*

$$\|g_\sigma\|_{H^2(\rho_*)} \geq \frac{C'}{\sqrt{\log(e+1/\sigma)}}.$$

*Proof.* For every odd $n \geq N(M)$, the cap remains below $q_*^n/4$. Since $a_n \geq q_*^n/2$,

$$|\tau_n - a_n| \geq \frac{1}{4} q_*^n.$$

After scaling by $\rho_*^n/n$, every such odd coefficient contributes at least $1/(4n)$ in modulus. Therefore

$$\|g\|^2_{H^2(\rho_*)} \geq \frac{1}{16} \sum_{j\geq 0} \frac{1}{(N+2j)^2} \geq \frac{1}{32N}.$$

The defining inequality gives $N(M) = \log M / \log(q_*/q) + O(1)$. Since $M \mapsto 1/\sqrt{\log(e+M)}$ is decreasing, the archived upper cap $M_\sigma \leq \sigma^{-1}$ gives the final statement; bounded masses only strengthen it. □

## 4 Protocol and exact boundary

The computation reports the exact forced odd cutoff and certified lower bound for $M = 10^4, 10^8, 10^{16}$, together with a separately named normalized logarithmic scale. Infinite model $\sum_{n>N} n^{-2}$ tails are represented by rigorous integral bounds rather than a finite truncation.

*Remark* 4.1. The lower bound tends to zero. It rules out endpoint $H^2$ decay faster than the displayed logarithmic scale but proves neither endpoint convergence nor divergence. The endpoint $H^\infty$ route is unavailable because the actual mismatch is not in that space. Gates A–E remain false/open.